

EmbedUR Linux Training

Module - 1 : Introduction to Linux OS,
Architecture and Kernel

Assessment - 1

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Question – 1: Create a file and add executable permission to all users (user, group and others)

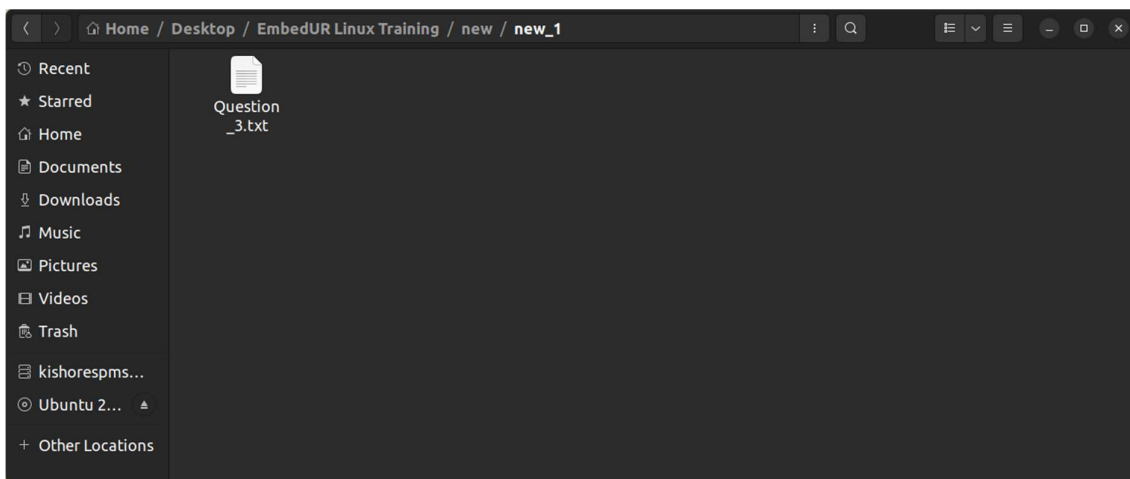
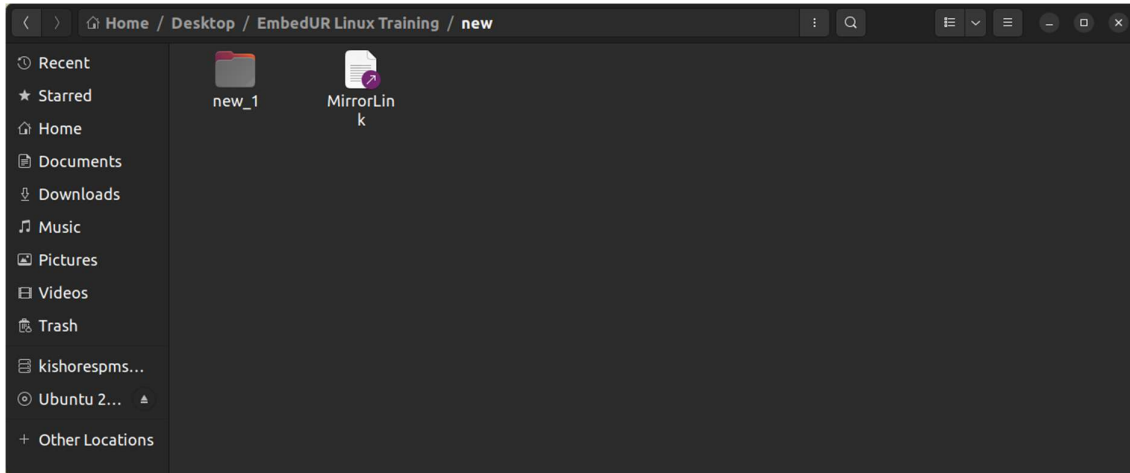
```
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Training
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ touch Question_1.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ ls -l
total 4
drwxrwxr-x 2 kishore kishore 4096 Jan 26 11:51 new
-rw-rw-r-- 1 kishore kishore 0 Jan 27 10:03 Question_1.txt
-rwxr-x--- 1 kishore kishore 0 Jan 27 09:29 Temp.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ chmod a+x Question_1.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ ls -l
total 4
drwxrwxr-x 2 kishore kishore 4096 Jan 26 11:51 new
-rwxrwxr-x 1 kishore kishore 0 Jan 27 10:03 Question_1.txt
-rwxr-x--- 1 kishore kishore 0 Jan 27 09:29 Temp.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$
```

Question – 2: Create a file and remove write permission for group user alone.

```
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Training
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ touch Question_2.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ ls -l
total 4
drwxrwxr-x 2 kishore kishore 4096 Jan 26 11:51 new
-rwxrwxr-x 1 kishore kishore 0 Jan 27 10:03 Question_1.txt
-rw-rw-r-- 1 kishore kishore 0 Jan 27 10:05 Question_2.txt
-rwxr-x--- 1 kishore kishore 0 Jan 27 09:29 Temp.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ chmod g-w Question_2.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ ls -l
total 4
drwxrwxr-x 2 kishore kishore 4096 Jan 26 11:51 new
-rwxrwxr-x 1 kishore kishore 0 Jan 27 10:03 Question_1.txt
-rw-r--r-- 1 kishore kishore 0 Jan 27 10:05 Question_2.txt
-rwxr-x--- 1 kishore kishore 0 Jan 27 09:29 Temp.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$
```

Question – 3 : Create a file and add a softlink to the file in different directory (Eg : Create a file in dir1/dir2/file and create a softlink for file inside dir1)

```
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Training/new
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new$ cd ../
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ mkdir new
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ cd new
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new$ mkdir new_1
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new$ cd new_1
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new/new_1$ ls
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new/new_1$ touch Question_3.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new/new_1$ ls -l
total 0
-rw-rw-r-- 1 kishore kishore 0 Jan 27 10:28 Question_3.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new/new_1$ cd ../
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new$ ln -s new_1/Question_3.txt MirrorLink
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new$ ls
MirrorLink new_1
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new$ ls -l
total 4
lrwxrwxrwx 1 kishore kishore 20 Jan 27 10:29 MirrorLink -> new_1/Question_3.txt
drwxrwxr-x 2 kishore kishore 4096 Jan 27 10:28 new_1
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/new$
```



Question - 4: Use ps command with options to display all active process running on the system

```
kishore@kishore-virtual-machine: ~/Desktop
kishore@kishore-virtual-machine:~/Desktop$ ps -ef
UID          PID    PPID  C  STIME TTY          TIME CMD
root         1        0  0  09:14 ?        00:00:03 /sbin/init splash
root         2        0  0  09:14 ?        00:00:00 [kthreadd]
root         3        2  0  09:14 ?        00:00:00 [pool_workqueue_release]
root         4        2  0  09:14 ?        00:00:00 [kworker/R-rcu_g]
root         5        2  0  09:14 ?        00:00:00 [kworker/R-rcu_p]
root         6        2  0  09:14 ?        00:00:00 [kworker/R-slub_]
root         7        2  0  09:14 ?        00:00:00 [kworker/R-netns]
root        11        2  0  09:14 ?        00:00:00 [kworker/u256:0-ext4-rsv-conversion]
root        12        2  0  09:14 ?        00:00:00 [kworker/R-mm_pe]
root        13        2  0  09:14 ?        00:00:00 [rcu_tasks_kthread]
root        14        2  0  09:14 ?        00:00:00 [rcu_tasks_rude_kthread]
root        15        2  0  09:14 ?        00:00:00 [rcu_tasks_trace_kthread]
root        16        2  0  09:14 ?        00:00:00 [ksoftirqd/0]
root        17        2  0  09:14 ?        00:00:02 [rcu_preempt]
root        18        2  0  09:14 ?        00:00:00 [migration/0]
root        19        2  0  09:14 ?        00:00:00 [idle_inject/0]
root        20        2  0  09:14 ?        00:00:00 [cpuhp/0]
root        21        2  0  09:14 ?        00:00:00 [cpuhp/1]
root        22        2  0  09:14 ?        00:00:00 [idle_inject/1]
root        23        2  0  09:14 ?        00:00:00 [migration/1]
root        24        2  0  09:14 ?        00:00:00 [ksoftirqd/1]
root        29        2  0  09:14 ?        00:00:00 [kdevtmpfs]
root        30        2  0  09:14 ?        00:00:00 [kworker/R-inet_]
root        31        2  0  09:14 ?        00:00:00 [kauditd]
root        33        2  0  09:14 ?        00:00:00 [khungtaskd]
root        34        2  0  09:14 ?        00:00:00 [oom_reaper]
```

```
kishore@kishore-virtual-machine: ~/Desktop

kishore 5713 1449 0 10:44 ? 00:00:00 /usr/bin/seahorse --gaplication-service
kishore 5841 1567 0 10:45 ? 00:00:00 /usr/libexec/gvfsd-network --spawner :1.13 /org/gtk/gvfs/exec_spaw/1
kishore 5855 1567 0 10:45 ? 00:00:00 /usr/libexec/gvfsd-dnssd --spawner :1.13 /org/gtk/gvfs/exec_spaw/3
root 6128 2 0 10:49 ? 00:00:00 [kworker/u258:3-events_unbound]
root 6615 2 0 10:59 ? 00:00:00 [kworker/u257:4-events_unbound]
root 6632 2 0 11:03 ? 00:00:00 [kworker/0:2-rcu_par_gp]
root 6703 2 0 11:04 ? 00:00:00 [kworker/1:1-events]
root 6865 2 0 11:09 ? 00:00:00 [kworker/u260:4-ttm]
root 6960 2 0 11:14 ? 00:00:00 [kworker/u260:0-ttm]
root 6968 2 0 11:14 ? 00:00:00 [kworker/0:0-mm_percpu_wq]
root 7038 2 0 11:14 ? 00:00:00 [kworker/u257:0-events_power_efficient]
kishore 7039 1449 0 11:14 ? 00:00:00 /usr/bin/python3 /usr/bin/gnome-terminal --wait
kishore 7040 7039 0 11:14 ? 00:00:00 /usr/bin/gnome-terminal.real --wait
kishore 7045 1449 0 11:14 ? 00:00:01 /usr/libexec/gnome-terminal-server
kishore 7075 7045 0 11:14 pts/0 00:00:00 bash
root 7088 2 0 11:14 ? 00:00:00 [kworker/u261:1-ttm]
root 7093 2 0 11:14 ? 00:00:00 [kworker/u261:8-ttm]
root 7105 2 0 11:15 ? 00:00:00 [kworker/1:0-events]
root 7108 2 0 11:15 ? 00:00:00 [kworker/u258:1-events_unbound]
root 7117 2 0 11:20 ? 00:00:00 [kworker/u261:0-ttm]
root 7118 2 0 11:20 ? 00:00:00 [kworker/u261:2]
root 7119 2 0 11:22 ? 00:00:00 [kworker/u257:2-flush-8:0]
root 7123 1 0 11:25 ? 00:00:00 /usr/libexec/fprintd
kishore 7137 1449 2 11:25 ? 00:00:00 /usr/bin/nautilus --gaplication-service
kishore 7145 1607 3 11:25 ? 00:00:00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/ding.js -E -P /u
kishore 7175 1449 2 11:25 ? 00:00:00 /usr/bin/file-roller --service
kishore 7200 7075 0 11:25 pts/0 00:00:00 ps -ef
kishore@kishore-virtual-machine: ~/Desktop$
```

Question – 5: Create 3 files in a dir1 and re-direct the output of list command with sorted by timestamp of the files to a file

```
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Training/Question_5

kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ mkdir Question_5
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training$ cd Question_5
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$ touch File_1.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$ touch File_2.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$ touch File_3.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$ ls
File_1.txt File_2.txt File_3.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$ ls | sort > Output.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$ cat Output.txt
File_1.txt
File_2.txt
File_3.txt
Output.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$ ls -lt > Output.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$ cat Output.txt
total 0
-rw-rw-r-- 1 kishore kishore 0 Jan 27 11:33 Output.txt
-rw-rw-r-- 1 kishore kishore 0 Jan 27 11:31 File_3.txt
-rw-rw-r-- 1 kishore kishore 0 Jan 27 11:30 File_2.txt
-rw-rw-r-- 1 kishore kishore 0 Jan 27 11:30 File_1.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$ ls -lt | sort > Output.txt
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$ cat Output.txt
-rw-rw-r-- 1 kishore kishore 0 Jan 27 11:30 File_1.txt
-rw-rw-r-- 1 kishore kishore 0 Jan 27 11:30 File_2.txt
-rw-rw-r-- 1 kishore kishore 0 Jan 27 11:31 File_3.txt
-rw-rw-r-- 1 kishore kishore 0 Jan 27 11:34 Output.txt
total 0
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Training/Question_5$
```